

Q63: the vanishing branch.
A closed form for form (ii) of $[e_i, R^{(\ell)}(t)]$,
and the proof that it is empty at level one

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Abstract

The 2 September paper proved that every nonzero entry of $[e_i, R^{(\ell)}(t)]$ is either $\varepsilon q^k t^b (1 + q^j t)$ with j known in closed form (form (i)), or $t^h (q^{-N_1} - q^{-N_2})$ (form (ii)), and left $N_1 - N_2$ uncomputed — “nobody has looked”. We look.

The census first. All 1604 form-(ii) entries of the standing sweep have exactly two contributing paths, in *opposite* commutator terms, with equal ribbon heights. They split into three structural cases, and the split is the theorem: (A) the ribbon and the node sit on different runners; (B) same runner, two beads moved; (C) same runner, one bead moved, the two paths lying in opposite terms.

Then the closed form, in three parts. Case (B) **never occurs**: $N_1 = N_2$ identically, so the entry is zero. Case (A) gives $N_1 - N_2 = \pm 1$, read off a *two-site indicator* comparing the node’s content with an end of the ribbon, which end being selected by the tie-break order of the two runners. Case (C) is governed by the σ/τ vocabulary of Theorem **thm:j**:

$$N_1 - N_2 = -\varepsilon(\sigma(\lambda; c, d) + \tau(\lambda; c + e, d)), \quad \varepsilon = +1 \iff c - 1 \in M_d(\lambda)$$

— the same two half-tie-terms, at the same two contents, with the leading 1 of Theorem **thm:j** absent and the sign reversed. So the brief’s instruction to test the surviving vocabulary before inventing a new one was right for exactly half the branch, and I say below which half and why.

Everything is proved from two lemmas: a *ribbon-shift lemma* computing how a Chevalley exponent changes when an e -ribbon is inserted underneath it (this does cases (A) and (B)), and the telescope-difference computation of 2 September rerun with the two paths in opposite terms (this does case (C)).

Gap 2 closes. Form (ii) is *empty at* $\ell = 1$, proved, not observed: case (A) needs two runners, case (B) cancels always, and case (C) has $\sigma = \tau = 0$ at $\ell = 1$. This is a cancellation and not a vacuity — at $\ell = 1$ there are 1424 live opposite-term path pairs in the checked range, every one of which sums to zero. The node **Q63-formii-empty-at-level-1** is promoted from **computed** to **proved**, and with it the classification of $[e_i, R^{(\ell)}(t)]$ is complete.

Verification: 1604/1604 in sample; 1564/1564 out of sample over seven configurations, including $\ell = 4$, which exhibits $|N_1 - N_2| = 3$, a value outside every in-sample range and inside the one now proved. Thirteen negative controls, each checked non-degenerate before being scored, all fail; the floor (the constant $N_1 - N_2 \equiv 1$) is 1530/3168.

Scope, stated here and not in a closing remark. Proposition **prop:heis** of 31 August shows $R^{(\ell)}(-q^{-1}) \neq B_{-1}^{[\ell]}$ for $\ell \geq 2$. What is classified below is therefore the commutator of a concretely defined graded ribbon operator, and *not* known to be a statement about the level- ℓ Heisenberg action on Fock space. Nothing in this paper narrows that gap.

1 What was open, and what closes

Theorem `thm:dich` of 2 September classified every nonzero entry $\Phi_{\nu\lambda}(t)$ of $[e_i, R^{(\ell)}(t)]$ into

- **form (i):** $\varepsilon q^k t^b (1 + q^j t)$, the two paths in the same commutator term, with $j = 1 + \varepsilon(\sigma(\lambda; c, d) + \tau(\lambda; c + e, d))$ (Theorem `thm:j`);
- **form (ii):** $t^h (q^{-N_1} - q^{-N_2})$, the two paths in opposite terms, vanishing at $q = 1$, with $N_1 - N_2$ *uncomputed*.

Gaps 1 and 2 of that paper were: compute $N_1 - N_2$; and decide whether form (ii) is empty at $\ell = 1$, where a 350-entry sweep had found none but Theorem `thm:dich` permitted them. Both close here, and they close together, because the closed form is what decides the emptiness.

Order of work, and an honest note about it. The brief asked for the census before any formula. The census was run (§3) and is reported first. But I must record that I did not arrive at the formulas by looking at the census: reading the proof of Lemma `lem:paths` and of Theorem `thm:j` on the way in, I saw that the same two computations answer the other branch, and I wrote down the three predictions of §4 *before* running `census.py`. That is a stronger position than fitting, not a weaker one — the census then tests a pre-registered prediction rather than confirming a description of itself — but it is a different position from the one the brief specified, and the reader should know which they are being offered. The census still did work the prediction could not: it is what shows that case (B) pairs are *numerous* rather than absent (§3), which is the difference between a theorem and a vacuity.

2 Conventions

Those of 31 August and 2 September, unchanged: rank $n = e \geq 2$, level $\ell \geq 1$; $\lambda = (\lambda^{(1)}, \dots, \lambda^{(\ell)})$ with multicharge $\mathbf{s}$; runner- d shifted Maya set $M_d = \{\lambda_j^{(d)} - j + s_d\}$, with the deep tail (all positions below the stored profile occupied) and the far top (none occupied). A node of component d at shifted content x is the bead move $x - 1 \leftrightarrow x$ on runner d ; an e -ribbon is the bead move $b \mapsto b + e$, of height $h = \#(M_d \cap (b, b + e))$. The tie-break is $\prec$ with the standard sign: (x, d') is *below* (x, d) iff $d' > d$. Write $d' \prec d$ for “runner d' is below runner d ”.

Throughout,

$$\chi_d(x) = [\text{addable at } (d, x)] - [\text{removable at } (d, x)].$$

Lemma 2.1 (The abacus form of χ). $\chi_d(x) = \mathbf{1}_{M_d}(x - 1) - \mathbf{1}_{M_d}(x)$.

Proof. Addable at (d, x) means $x - 1 \in M_d$, $x \notin M_d$; removable means $x - 1 \notin M_d$, $x \in M_d$. Writing $a = \mathbf{1}_{M_d}(x - 1)$, $b = \mathbf{1}_{M_d}(x)$, the two indicators are $a(1 - b)$ and $b(1 - a)$, whose difference is $a - b$. $\square$

This one-line identity is what makes everything below finite and local: χ_d is a difference of indicators, so it telescopes, and it is supported on the finitely many positions where M_d is not constant.

Definition 2.2. For $x \in \mathbb{Z}$ and a runner d ,

$$\tau(\lambda; x, d) = \sum_{d' \prec d} \chi_{d'}(x), \quad \sigma(\lambda; x, d) = \sum_{d' \succ d} \chi_{d'}(x), \quad A_{d'}(x) = \sum_{k \geq 1} \chi_{d'}(x - ke),$$

so that Theorem `thm:tele` reads $N^<(\lambda, \gamma) = \sum_{d'} A_{d'}(x) + \tau(\lambda; x, d)$ for a node γ at (d, x) . Both σ and τ read only runners $\neq d$, so both are unchanged if λ is replaced by any multipartition agreeing with it off runner d . At $\ell = 1$ both are empty sums, hence 0.

Recall from Definition `def:chev` that the e_i weight is $q^{-N^<}$ with $N^<$ read on the *target* of the removal. A *path* is a ribbon move together with a node removal, in one of the two orders; the $e_i R$ order contributes with sign +1 and the Re_i order with sign -1. For a form-(ii) entry with one path in each term we always write N_1 for the exponent of the $e_i R$ path and N_2 for that of the Re_i path, so that $\Phi = t^h(q^{-N_1} - q^{-N_2})$.

3 The census

Run over the nine triples $(e, \ell, \mathbf{s})$ of Remark `rem:verif-rigid`, all multipartitions of size ≤ 4 , all i ; bead depth $|\lambda| + 2e + 6$. The same run reproduces 1248 form-(i) entries and 0 unclassified entries, which is the check that this session's pipeline computes the object of the previous two sessions.

Proposition 3.1 (Form-(ii) census). *Over all 1604 form-(ii) entries: every entry has exactly two contributing paths; in every entry the two paths lie in opposite commutator terms (1604/1604); in every entry the two ribbon heights are equal (1604/1604); and every entry falls into exactly one of*

case	description	count	$N_1 - N_2$ observed
(A)	ribbon and node on different runners	1146	± 1 only
(B)	one runner, two beads moved	0	—
(C)	one runner, one bead moved, opposite terms	458	$\pm 1, \pm 2$

In cases (A) and (B) the two paths use the same ribbon and the same node, in the two orders. In case (C) they use different ribbons and different nodes: the c -path and the $(c + e)$ -path of Lemma `lem:paths`.

Three things in this table are worth naming before they are proved.

The opposite-term fact is forced, and is the mirror of the form-(i) one. A form-(ii) entry is a single t -power whose coefficient vanishes at $q = 1$, i.e. a difference of two equal-signed monomials appearing with opposite signs; the two terms of a commutator enter with opposite signs. (For form (i) the 2 September paper ran the same argument the other way.)

Case (B) is empty, and that is a claim about entries, not about pairs. An entry-level census cannot distinguish “case (B) always cancels” from “case (B) never happens”. *Variation exercised:* enumerate path *pairs* rather than entries, keeping those whose sum is zero, and ask separately whether case-(B) pairs occur and whether they cancel.

configuration	case-(B) pairs	nonzero	$N_1 - N_2$
$e=2, \ell=2, \lambda \leq 4$	60	0	0 throughout
$e=3, \ell=2, \lambda \leq 4$	94	0	0 throughout
$e=5, \ell=2, \lambda \leq 3$	72	0	0 throughout
$e=2, \ell=3, \lambda \leq 3$	36	0	0 throughout
$e=3, \ell=3, \lambda \leq 3$	66	0	0 throughout
$e=2, \ell=4, \lambda \leq 3$	56	0	0 throughout

So case-(B) pairs are abundant — 384 of them here — and every one of them cancels. The emptiness is a theorem about a live population.

The case-(C) values reach $\ell - 1$. At $\ell = 2$, $|N_1 - N_2| \leq 1$; at $\ell = 3$ the value 2 appears (24 times); the out-of-sample $\ell = 4$ run produces 3. Case (A) never exceeds 1 at any level. That is the shape the theorem below predicts, and it is the first sign that (A) and (C) are governed by different statistics.

Census question 2: which object does each exponent read from? N_1 is read on the target ν of the entry; N_2 is read on the intermediate $\rho = \lambda - \gamma$, which is neither λ nor ν . Their *difference*, by Theorems 4.3 and 4.6 below, is a function of λ and the net bead displacement alone — intrinsic, exactly as j turned out to be.

Census question 3: are N_1 and N_2 separately meaningful? Yes, in the weak sense that each is a well-defined finite integer: *variation exercised*, the abacus depth raised by 6; over 590 tracked pairs neither N_1 nor N_2 nor their difference moved, so none of the three is a truncation artefact. But only the difference has a closed form of bounded size: over the full sweep N_1 ranges over $[-3, 5]$ and N_2 over $[-2, 5]$, growing with $|\lambda|$ through the unbounded telescope $\sum_{d'} A_{d'}$, while $N_1 - N_2$ is confined to $[-3, 3]$. I therefore write no formula for N_1 alone, as the brief instructed.

4 The closed form

4.1 Cases (A) and (B): the ribbon-shift lemma

Everything about the same-moves cases follows from one computation: what an e -ribbon does to a Chevalley exponent sitting elsewhere.

Lemma 4.1 (Ribbon shift). *Let ρ be a multipartition, γ a node at (d_0, x_0) addable in ρ , and let the ribbon move $b \mapsto b + e$ on runner d_1 be legal in ρ ; write $\tilde{\rho}$ for its result. Then*

$$N^<(\tilde{\rho}, \gamma) - N^<(\rho, \gamma) = (\delta_{x_0, b+e} - \delta_{x_0, b+e+1}) + [d_1 \prec d_0](\delta_{x_0, b} - \delta_{x_0, b+1} - \delta_{x_0, b+e} + \delta_{x_0, b+e+1}).$$

Proof. By Theorem `thm:tele`, $N^<(\rho, \gamma) = \sum_{d'} A_{d'}(x_0) + \sum_{d' \prec d_0} \chi_{d'}(x_0)$, and only runner d_1 changes. There $\mathbf{1}_{\tilde{M}} - \mathbf{1}_M = -\delta_b + \delta_{b+e}$, so by Lemma 2.1

$$\Delta \chi_{d_1}(y) = (\mathbf{1}_{\tilde{M}} - \mathbf{1}_M)(y - 1) - (\mathbf{1}_{\tilde{M}} - \mathbf{1}_M)(y) = \delta_{y, b} - \delta_{y, b+1} - \delta_{y, b+e} + \delta_{y, b+e+1}.$$

The four sites lie in two residue classes: $b, b + e \equiv b$ and $b + 1, b + e + 1 \equiv b + 1 \pmod{e}$. Now $\Delta A_{d_1}(x_0) = \sum_{k \geq 1} \Delta \chi_{d_1}(x_0 - ke)$, a sum over $x_0 - e, x_0 - 2e, \dots$. If $x_0 \equiv b$, the term $+\delta_{y, b}$ contributes

$+1$ exactly when $x_0 \geq b+e$ and the term $-\delta_{y,b+e}$ contributes -1 exactly when $x_0 \geq b+2e$; the net is $\delta_{x_0,b+e}$. If $x_0 \equiv b+1$, the terms $-\delta_{y,b+1}$ and $+\delta_{y,b+e+1}$ contribute -1 when $x_0 \geq b+e+1$ and $+1$ when $x_0 \geq b+2e+1$; the net is $-\delta_{x_0,b+e+1}$. Otherwise both classes miss and $\Delta A_{d_1}(x_0) = 0$. In all cases $\Delta A_{d_1}(x_0) = \delta_{x_0,b+e} - \delta_{x_0,b+e+1}$. The tie term contributes $\Delta \chi_{d_1}(x_0)$ if and only if $d_1 \prec d_0$ (in particular never when $d_1 = d_0$, since $\prec$ is strict). Adding the two gives the statement. $\square$

Remark 4.2. The lemma says something one could have guessed the shape of but not the support: an e -ribbon is invisible to every Chevalley exponent except at the four sites $b, b+1, b+e, b+e+1$, and the telescope sees only the *upper* pair. A ribbon inserted far below a node does not perturb its weight at all, however many beads it passes. That is why the answer is a two-site indicator and not a count.

Theorem 4.3 (Case (A)). *Let $\Phi_{\nu\lambda}$ be an entry whose net bead displacement moves one bead $b \mapsto b+e$ on runner d_1 and one bead $x_0 \mapsto x_0-1$ on a runner $d_0 \neq d_1$. Then it has exactly two paths, in opposite terms, with equal ribbon heights, and*

$$N_1 - N_2 = \begin{cases} \delta_{x_0,b} - \delta_{x_0,b+1}, & d_1 \prec d_0, \\ \delta_{x_0,b+e} - \delta_{x_0,b+e+1}, & d_1 \succ d_0. \end{cases}$$

In particular $N_1 - N_2 \in \{-1, 0, +1\}$, and the entry is nonzero exactly when x_0 is one of the two indicated sites — the lower end of the ribbon (and its successor) if the ribbon lies on the lower runner, the upper end (and its successor) if on the upper.

Proof. The two moves are on different runners, hence independent: both orders are legal, the pair is determined by the net displacement (a ribbon moves a bead by $+e$ and a node by -1 , and $e \geq 2$), and there are no other paths. Put $\rho = \lambda - \gamma$. In the $e_i R$ order the ribbon acts on λ and the removal has target ν ; in the Re_i order the removal has target ρ and the ribbon then carries ρ to ν . Hence $\nu = \tilde{\rho}$ and

$$N_1 = N^<(\tilde{\rho}, \gamma), \quad N_2 = N^<(\rho, \gamma).$$

The heights are equal because the ribbon window lies on runner d_1 and the node move is on $d_0 \neq d_1$, so M_{d_1} is the same in λ and in ρ . Lemma 4.1 with base ρ gives the difference; when $d_1 \prec d_0$ the two brackets combine to $\delta_{x_0,b} - \delta_{x_0,b+1}$, and when $d_1 \succ d_0$ only the first survives. All the data b, d_1 may be read in λ , since λ and ρ agree on runner d_1 . $\square$

Theorem 4.4 (Case (B): the entry is always zero). *Let $\Phi_{\nu\lambda}$ be an entry whose net displacement moves two beads on a single runner d , by $b \mapsto b+e$ and $x_0 \mapsto x_0-1$ with $x_0 \neq b+e$ and $x_0-1 \neq b$. Then $h_1 = h_2$ and $N_1 = N_2$, so $\Phi_{\nu\lambda} = 0$. No form-(ii) entry arises this way.*

Proof. Both orders are legal, and legality is more restrictive than it looks. Write $M = M_d(\lambda)$. In the $e_i R$ order the ribbon needs $b \in M, b+e \notin M$; then removal at x_0 needs $x_0 \in \tilde{M}$ and $x_0-1 \notin \tilde{M}$ where $\tilde{M} = M \setminus \{b\} \cup \{b+e\}$, which given $x_0 \neq b+e$ and $x_0-1 \neq b$ forces $x_0 \in M, x_0 \neq b, x_0-1 \notin M, x_0-1 \neq b+e$. So

$$x_0 \notin \{b, b+1, b+e, b+e+1\},$$

the last two by hypothesis and the first two by legality. This is precisely the support of Lemma 4.1, and since $d_1 = d_0 = d$ the bracket $[d_1 \prec d_0]$ vanishes; hence $N_1 - N_2 = \delta_{x_0,b+e} - \delta_{x_0,b+e+1} = 0$.

Heights: $h_1 = \#(M \cap (b, b+e))$ and h_2 is the same count for $M' = M \setminus \{x_0\} \cup \{x_0 - 1\}$. The two differ only if exactly one of $x_0 - 1, x_0$ lies in $\{b+1, \dots, b+e-1\}$, i.e. only if $x_0 = b+1$ or $x_0 = b+e$, both excluded. So the two monomials $+t^{h_1}q^{-N_1}$ and $-t^{h_2}q^{-N_2}$ are equal and opposite. $\square$

Remark 4.5. Theorem 4.4 is the pleasant kind of result: the four sites at which a ribbon can perturb a Chevalley exponent are exactly the four sites at which the node move would collide with the ribbon and so are exactly the four forbidden by legality. The two moves are far enough apart to commute precisely because they are far enough apart to both be legal. The census (§3) confirms the mechanism directly: over the six configurations tabulated there the offset $x_0 - b$ takes the values $\{-6, \dots, -1\} \cup \{2, \dots, e-1\} \cup \{e+2, \dots\}$ and *never* lands in $\{0, 1, e, e+1\}$.

4.2 Case (C): the σ/τ vocabulary, again

Theorem 4.6 (Case (C)). *Let $\Phi_{\nu\lambda}$ be an entry whose net displacement moves a single bead $c \mapsto c+e-1$ on runner d , with the two paths of Lemma **lem:paths** lying in opposite terms — equivalently, with $(c-1 \in M_d(\lambda)) \Leftrightarrow (c+e \in M_d(\lambda))$. Then $h_1 = h_2$ and*

$$N_1 - N_2 = -\varepsilon(\sigma(\lambda; c, d) + \tau(\lambda; c+e, d)), \quad \varepsilon = \begin{cases} +1, & c-1 \in M_d(\lambda), \\ -1, & c-1 \notin M_d(\lambda). \end{cases}$$

Consequently $|N_1 - N_2| \leq \ell - 1$, and the entry is nonzero exactly when $\sigma(\lambda; c, d) + \tau(\lambda; c+e, d) \neq 0$.

Proof. Write $M = M_d(\lambda)$; by hypothesis $c \in M$ and $c+e-1 \notin M$. By Lemma **lem:paths** the two paths are the c -path (ribbon $c-1 \mapsto c+e-1$, node at content c) and the $(c+e)$ -path (ribbon $c \mapsto c+e$, node at content $c+e$), and the c -path lies in $e_i R$ iff $c-1 \in M$ while the $(c+e)$ -path lies in $e_i R$ iff $c+e \notin M$. Opposite terms is therefore the stated equivalence, and ε is the sign of the case as in Theorem **thm:j**.

Heights. With $H = \#(M \cap \{c+1, \dots, c+e-2\})$, the height table of Lemma **lem:paths** gives, for $\varepsilon = +1$, $h_c = H+1$ (c -path in $e_i R$) and $h_{c+e} = H+1$ ($(c+e)$ -path in Re_i); for $\varepsilon = -1$, $h_c = H$ and $h_{c+e} = H$. Equal in both cases, so the entry is a single t -power.

Exponents. Set

$$D := N^<(\text{base of the } (c+e)\text{-path}, \gamma_{c+e}) - N^<(\text{base of the } c\text{-path}, \gamma_c),$$

each base being the target of that path's removal. Since for $\varepsilon = +1$ the c -path is the $e_i R$ one and for $\varepsilon = -1$ it is the Re_i one, in both cases $N_1 - N_2 = -\varepsilon D$. It remains to show $D = \sigma(\lambda; c, d) + \tau(\lambda; c+e, d)$.

Both bases agree with λ off runner d . Off runner d , using $A_{d'}(x+e) = \chi_{d'}(x) + A_{d'}(x)$,

$$\sum_{d' \neq d} (A_{d'}(c+e) - A_{d'}(c)) = \sum_{d' \neq d} \chi_{d'}(c) = \sigma(\lambda; c, d) + \tau(\lambda; c, d),$$

and the two tie terms contribute $\tau(\lambda; c+e, d) - \tau(\lambda; c, d)$; together $\sigma(\lambda; c, d) + \tau(\lambda; c+e, d)$, as required. So it suffices to show that the runner- d contribution to D vanishes.

Case $\varepsilon = +1$ ($c-1 \in M, c+e \in M$). The c -path's base is ν with $M_d(\nu) = M \setminus \{c\} \cup \{c+e-1\}$; the $(c+e)$ -path's base is ρ with $M_d(\rho) = M \setminus \{c+e\} \cup \{c+e-1\}$. The runner- d contribution is $A_d^\rho(c+e) - A_d^\nu(c)$.

- $A_d^\nu(c)$ has arguments $c-e, c-2e, \dots$, all $\leq c-2$. Now ν differs from M only at c and $c+e-1$, so χ_d^ν differs from χ_d^M only at $\{c, c+1, c+e-1, c+e\}$, all $\geq c$. Hence $A_d^\nu(c) = A_d^M(c)$.
- $A_d^\rho(c+e) = \chi_d^\rho(c) + A_d^\rho(c)$. Here ρ differs from M only at $c+e-1, c+e$, so χ_d^ρ differs from χ_d^M only at $\{c+e-1, c+e, c+e+1\}$, all $> c$; hence $A_d^\rho(c) = A_d^M(c)$ and $\chi_d^\rho(c) = \chi_d^M(c) = \mathbf{1}_M(c-1) - \mathbf{1}_M(c) = 1 - 1 = 0$, using $c-1 \in M$ and $c \in M$.

So the runner- d contribution is $A_d^M(c) - A_d^M(c) = 0$.

Case $\varepsilon = -1$ ($c-1 \notin M, c+e \notin M$). The $(c+e)$ -path's base is ν with $M_d(\nu) = M \setminus \{c\} \cup \{c+e-1\}$; the c -path's base is ρ' with $M_d(\rho') = M \setminus \{c\} \cup \{c-1\}$. The runner- d contribution is $A_d^\nu(c+e) - A_d^{\rho'}(c)$.

- $A_d^\nu(c+e) = \chi_d^\nu(c) + A_d^\nu(c) = \chi_d^\nu(c) + A_d^M(c)$ by the first bullet above, and $\chi_d^\nu(c) = \mathbf{1}_\nu(c-1) - \mathbf{1}_\nu(c) = 0 - 0 = 0$, using $c-1 \notin M$ (unchanged in ν) and $c \notin M_d(\nu)$.
- $A_d^{\rho'}(c)$ has arguments $\leq c-e \leq c-2$. Now ρ' differs from M only at $c-1, c$, so $\chi_d^{\rho'}$ differs from χ_d^M only at $\{c-1, c, c+1\}$, all $\geq c-1 > c-2$. Hence $A_d^{\rho'}(c) = A_d^M(c)$.

Again the runner- d contribution is 0. In both cases $D = \sigma(\boldsymbol{\lambda}; c, d) + \tau(\boldsymbol{\lambda}; c+e, d)$.

The bound: $\sigma(\boldsymbol{\lambda}; c, d)$ is a sum of $\chi_{d'}(c) \in \{-1, 0, 1\}$ over the runners above d and $\tau(\boldsymbol{\lambda}; c+e, d)$ a sum over those below, so $|\sigma + \tau| \leq \ell - 1$. $\square$

Remark 4.7 (What the vocabulary did and did not carry). The brief asked that the σ/τ vocabulary of Theorem **thm:j** be tested before any new statistic was invented. It carries case (C) exactly — same two half-tie terms, same two contents c and $c+e$, same bead condition $c-1 \in M_d$ selecting the sign — and it does not carry case (A) at all. The reason is structural rather than accidental: σ and τ measure cross-runner traffic *at the two ends of one ribbon window*, which is a quantity only the one-runner cases have. In case (A) the node and the ribbon are on different runners and there is no window to cross; what survives is the bare two-site indicator of Lemma 4.1. So the honest report is: the surviving vocabulary covered half the branch, and I can say which half and why.

Remark 4.8 (The relation to j). Comparing with Theorem **thm:j**, $j = 1 + \varepsilon(\sigma + \tau)$ and, in case (C), $N_1 - N_2 = -\varepsilon(\sigma + \tau)$, so on the same displacement data

$$j + (N_1 - N_2) = 1.$$

The two branches are the same computation with the two paths' terms swapped: swapping the terms reverses the sign of the second monomial and removes the leading 1, which is the runner- d term $\chi_d(c) = \pm 1$ of Theorem **thm:j** and is exactly what the two vanishing runner- d computations above replace by 0. Form (i) and form (ii) are not two phenomena; they are one formula read on the two sides of a bead condition.

5 Gap 2: form (ii) is empty at level one

Theorem 5.1. *At $\ell = 1$ every nonzero entry of $[e_i, R^{(1)}(t)]$ is of form (i), with $j = 1$. Form (ii) is empty.*

Proof. A path consists of one ribbon and one node move, so the net displacement Δ moves at most two beads, and the cases of Theorem **thm:dich** are exhaustive: Δ moves beads on two runners

(case (A)), or two beads on one runner (case (B)), or one bead on one runner (form (i) or case (C)). ($\Delta = 0$ is impossible: a ribbon displaces a bead by $+e$ and a node by -1 , and these cancel only if $e = 1$.)

At $\ell = 1$: case (A) cannot occur, there being one runner. Case (B) gives $N_1 = N_2$ by Theorem 4.4, at any level, so the entry is zero. Case (C) has $\sigma(\lambda; c, d) = \tau(\lambda; c + e, d) = 0$, both being empty sums over the runners other than d , so $N_1 - N_2 = 0$ by Theorem 4.6 and the entry is again zero. Hence every nonzero entry has its two paths in the same term, which is form (i), and there $j = 1 + \varepsilon \cdot 0 = 1$. $\square$

Remark 5.2 (Not a vacuity). *Variation exercised*: at $\ell = 1$, do case-(B) and case-(C) path pairs occur at all? If they did not, Theorem 5.1 would be true for a reason having nothing to do with the cancellation it claims. They occur in quantity:

configuration ($\ell = 1, \lambda \leq 7$)	case-(B) pairs	case-(C) pairs	total	nonzero
$e = 2, s = 0$	142	76	218	0
$e = 3, s = 0$	182	88	270	0
$e = 4, s = 0$	224	104	328	0
$e = 5, s = 0$	278	112	390	0
$e = 2, s = 1$	142	76	218	0
total	968	456	1424	0

All 1424 opposite-term pairs at $\ell = 1$ have $N_1 = N_2$ and $h_1 = h_2$, and every one of them cancels. The emptiness is a cancellation over a live population, which is what the theorem asserts.

Remark 5.3 (Independent confirmation, with a positive control). Separately, the full classification was rerun at $\ell = 1$ at sizes beyond every previous sweep — $(e, s, |\lambda|_{\max}) = (2, 0, 9), (3, 0, 8), (4, 0, 8), (5, 0, 8), (2, 3, 8)$, bead depths 18–24. It returns 607 form-(i) entries, 0 form-(ii) entries and 0 unclassified entries, with $j = 1$ on all 607. The 607 are the positive control: a run that found nothing at all would have proved nothing about form (ii).

6 The classification, complete

Theorem 6.1 (Classification of $[e_i, R^{(\ell)}(t)]$). *Let $\Phi_{\nu\lambda}(t) \neq 0$ be an entry of $[e_i, R^{(\ell)}(t)]$ and Δ the net change of Maya sets. Then $\Phi_{\nu\lambda}$ has exactly two contributing paths and exactly one of the following holds.*

- (i) Δ moves one bead $c \mapsto c + e - 1$ on a runner d , and $c - 1 \in M_d(\lambda) \Leftrightarrow c + e \notin M_d(\lambda)$. Then the two paths lie in the same term and $\Phi = \varepsilon q^k t^b (1 + q^j t)$ with $j = 1 + \varepsilon(\sigma(\lambda; c, d) + \tau(\lambda; c + e, d))$ and $\varepsilon = +1$ iff $c - 1 \in M_d(\lambda)$ (Theorem **thm:j**).
- (ii-C) Δ moves one bead $c \mapsto c + e - 1$ on a runner d , and $c - 1 \in M_d(\lambda) \Leftrightarrow c + e \in M_d(\lambda)$. Then the two paths lie in opposite terms and $\Phi = q^{-N_2} t^h (q^{-m} - 1)$ with $m = -\varepsilon(\sigma(\lambda; c, d) + \tau(\lambda; c + e, d)) \neq 0$; here $1 \leq |m| \leq \ell - 1$.
- (ii-A) Δ moves one bead $b \mapsto b + e$ on a runner d_1 and one bead $x_0 \mapsto x_0 - 1$ on a runner $d_0 \neq d_1$, with $x_0 = b$ or $b + 1$ if $d_1 \prec d_0$, and $x_0 = b + e$ or $b + e + 1$ if $d_1 \succ d_0$. Then $\Phi = q^{-N_2} t^h (q^{\mp 1} - 1)$, the sign being $-$ at the first of the two sites and $+$ at the second.

No other Δ gives a nonzero entry: two beads on one runner always cancel (Theorem 4.4), as does case (ii-C) with $\sigma + \tau = 0$ and case (ii-A) with x_0 at neither site. In particular (Theorem 5.1) at $\ell = 1$ only (i) survives, with $j = 1$.

Proof. Immediate from Theorem `thm:dich`, Theorem `thm:j`, and Theorems 4.3, 4.4, 4.6 above. $\square$

Corollary 6.2. *At $q = 1$ every form-(ii) entry vanishes and every form-(i) entry becomes $2\epsilon t^b$.*

Corollary 6.3 (Q59 recovered). *At $\ell = 1$ every nonzero entry of $[e_i, R^{(1)}(t)]$ equals $\epsilon q^k t^b (1 + qt)$.*

Proof. By Theorem 5.1 every nonzero entry is of form (i) with $j = 1$, and form (i) with $j = 1$ is $\epsilon q^k t^b (1 + qt)$. $\square$

This is Corollary `cor:Q59` of 31 August, whose content is that $(1 + qt)$ divides $[e_i, R_e(t)]$ entrywise, hence as a matrix: $[e_i, R_e(t)] = (1 + qt)D_i^-(t)$. So the level- ℓ classification degenerates to the level-1 rigidity theorem, as it must, and the divisibility fails at higher level for a reason the classification now names — $j \neq 1$ in case (i), and the form-(ii) entries, which are t -monomials and are divisible by $(1 + qt)$ only when they are zero.

7 Verification and negative controls

Same object. The nine in-sample triples return 1248 form-(i) entries and 0 unclassified entries, reproducing 2 September exactly; the 1604 form-(ii) entries are the complement.

In sample. Theorems 4.3 and 4.6: **1604/1604**.

Out of sample. Seven configurations absent from the in-sample list — $(e, \ell, \mathbf{s}) = (2, 2, (1, 0))$, $(3, 2, (0, 2))$, $(2, 4, (0, 0, 0, 0))$, $(5, 2, (0, 0))$, $(3, 3, (0, 1, 2))$, $(2, 2, (0, 0))$ at $|\lambda| \leq 5$, and $(2, 3, (0, 0, 0))$ at depth +6 — giving 1564 further form-(ii) entries. **1564/1564**. The $\ell = 4$ configuration produces $|N_1 - N_2| = 3$, a value no in-sample run had seen, predicted correctly; this is the form-(ii) analogue of the $j = 4$ event of 2 September.

Depth. The standing instruction is to check depth first on any failure. There were no failures, so the control was run positively: $(2, 3, (0, 0, 0))$ recomputed with the bead depth raised by 6 returns the same 354 form-(ii) entries with the same N_1 , N_2 and $N_1 - N_2$ throughout, and the separate depth-sensitivity run of §3 moved none of 590 tracked (N_1, N_2) pairs. The counts are not truncation artefacts.

Range. $|N_1 - N_2| \leq 1$ in case (A) at every level tested, and $|N_1 - N_2| \leq \ell - 1$ in case (C), *saturated* at each of $\ell = 2, 3, 4$.

Sign identification. All 3168 entries have coefficient exactly $q^{-N_1} - q^{-N_2}$ with N_1 from the $e_i R$ path, confirming the labelling on which the sign of the theorem depends.

Negative controls. Each was first checked for non-degeneracy — it must change at least one prediction — and the count of moved predictions is reported beside its score. *Every variant below moves a nonzero number of predictions; none is blind.* Scored on all 3168 entries.

variant	score	moved	variation exercised
Theorems 4.3, 4.6	3168/3168	—	—
(A) the two runner-order branches exchanged	872	2296	the order of d_1 vs d_0
(A) runner order ignored (always the $b + e$ branch)	2020	1148	whether d_1 vs d_0 matters
(A) the two-site sign flipped	872	2296	which site carries +1
(C) ε branch ignored	2732	436	the bead condition $c - 1 \in M_d$
(C) $\sigma \leftrightarrow \tau$ exchanged	2434	734	tie-break orientation
(C) both halves read at content c	2738	430	which content each half uses
(C) both halves read at content $c + e$	2738	430	which content each half uses
(C) the σ term dropped	2717	451	the runners above d
(C) the τ term dropped	2717	451	the runners below d
(C) overall sign flipped	2296	872	orientation of $e_i R - R e_i$
cross-case: (A)’s rule applied to (C) too	2296	872	the case split itself
cross-case: (C)’s rule applied to (A) too	872	2296	the case split itself
rival: sign read off the c -path’s term alone	1366	1802	a cheaper statistic
floor: the constant $N_1 - N_2 \equiv 1$	1530	1638	—

The floor is 1530/3168 (48.3%), which is what a constant can score on a sign-symmetric quantity. Every structural ingredient of the theorem is worth between 430 and 2296 moved predictions, and every perturbation of one loses.

Remark 7.1 (A control I wrote twice by mistake, and the correction). My first control list contained “ $\sigma \leftrightarrow \tau$ exchanged” and “both halves read at the other node’s content” as two entries. They are the same expression: sending $\sigma(c) + \tau(c + e) \mapsto \tau(c) + \sigma(c + e)$ is both operations at once. The table above replaces the second by two genuinely different variants (both halves at c , both at $c + e$). This is the same failure mode as the blind control of 2 September in a smaller key — a control that is not the control you named — and it is recorded rather than quietly fixed.

Planted errors. Perturbing 1, 5 and 20 of the *observed* $N_1 - N_2$ by +1 drops the score to 3167, 3163 and 3148: caught one for one, so the comparison can see a failure.

What else would pass. There is no fitted parameter. In case (A) the four sites are the support computed in Lemma 4.1 and the branch is the bracket $[d_1 \prec d_0]$ appearing in Theorem **thm:tele**; in case (C) σ and τ are the two halves of that same tie term, the contents c and $c + e$ are forced by Lemma **lem:paths**, and ε is the bead condition already proved to give the sign in Theorem **thm:j**. The one place a coincidence could hide is the case (A) branch, whose two halves are distinguished only on the 1148 entries with $d_1 \prec d_0$ — which is why the “runner order ignored” control is reported: it still scores 2020, and the branch is supported by the 1148 entries it moves.

8 Gaps

1. **Sizes.** $|\lambda| \leq 5$ at $\ell \leq 3$, ≤ 3 at $\ell = 4$; bead depths 12–24. $\ell \leq 4$, so σ and τ have been exercised with at most three other runners. The $\ell = 1$ runs reach $|\lambda| \leq 9$.
2. **One tie-break.** As on 2 September, everything is computed at tie-break +1. Under the opposite convention σ and τ exchange roles and so do the two branches of Theorem 4.3; the mirror statement is not written out. The $\sigma \leftrightarrow \tau$ control measures the difference and it is large (734 predictions), so the convention is doing real work and the mirror is not free.
3. **N_1 and N_2 separately.** Only the difference is given a closed form. N_1 alone is $\sum_{d'} A_{d'} + \tau$, an unbounded telescope; no shorter description of it is claimed or attempted. See §3, census question 3.
4. **Proposition `prop:heis` stands, untouched.** As stated in the abstract: for $\ell \geq 2$, $R^{(\ell)}(-q^{-1}) \neq B_{-1}^{[e]}$, so Theorem 6.1 classifies the commutator of a concretely defined operator and is not known to be a statement about the level- ℓ Heisenberg action. This is the programme’s largest open question and no session since 31 August has moved it.
5. **No representation-theoretic reading of the two form-(ii) cases.** The theorem says two-runner entries are always ± 1 and one-runner entries carry the cross-runner traffic, but I have no conceptual account of why the two-runner case is rigid. Lemma 4.1 explains it mechanically — a ribbon is invisible except at four sites — without saying what it means.

9 Corrections to earlier papers

1. Gap 1 of 2 September (“form (ii) is not computed”) is **closed** (Theorems 4.3, 4.4, 4.6).
2. Gap 2 of 2 September (“form (ii) is not proved empty at $\ell = 1$ ”) is **closed** (Theorem 5.1); the Corollary at the end of §`sec:dichotomy`, whose $\ell = 1$ emptiness clause was explicitly left computational, is now proved in full.
3. The 2 September speculation that “the same σ/τ analysis should apply” to form (ii) is **half right**: it applies verbatim to case (C) and not at all to case (A), which needs Lemma 4.1 instead. Recorded because the speculation was written as though the branch were homogeneous, and it is not.

Is this a paper?

Yes, with one named weakness. What now exists is a complete and self-contained classification of $[e_i, R^{(\ell)}(t)]$ for the graded ribbon operator $R^{(\ell)}(t)$ on level- ℓ Fock space: every nonzero entry is one of three explicit shapes, each shape’s exponent is given by a formula in the multipartition and the net bead displacement alone, every formula is proved from the telescope theorem rather than fitted, and the level-one case reduces to the Q59 rigidity statement as it must. It rests on three theorems proved across three sessions (`thm:tele`, `thm:j`, Theorem 6.1) with no remaining computational premises, it has 3168 out-of-sample-inclusive confirmations and a floor to measure

them against, and its negative controls are non-degenerate by construction. A referee could read it.

The single weakest point is not in the mathematics: it is the motivation. Proposition `prop:heis` says that for $\ell \geq 2$ the operator being classified is not the one the programme wanted — $R^{(\ell)}(-q^{-1})$ is not $B_{-1}^{[e]}$ — so a reader who does not already care about $R^{(\ell)}(t)$ has, at present, no reason given here to start. At $\ell = 1$ the operator *is* the right one and the paper’s level-one content is a genuine statement about the Heisenberg action; at higher level the paper is currently a theorem about a well-defined combinatorial operator whose representation theory is unestablished. Anyone deciding whether to promote this to `publishable-result` should decide that question first, because it determines whether the paper is “the commutator of the level- ℓ ribbon operator, classified” or “a level- ℓ analogue of Q59, with the analogy unverified”. The honest framing is the second, and the second is still worth publishing — but it must be the title’s framing, not a remark’s.